

SIMATS ENGINEERING

ASSESSMENT TOOL1

COURSE CODE: MLA 02 COURSE NAME: Fundamentals of Machine Learning

COURSE OUTCOME COVERED

CO2: *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 10 Total Marks:50 Annexure A

Analytical Problem Solving

Code of Conduct:

*I **SHAIK ARSHAD(192425080)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary

*action. **Signature of the student:ARSHAD***

Annexure A

Analytical Problem Solving

Q1.A real estate company collected the following training data:(15Marks -20

Mins) **House No Area (sq.ft) Bedrooms Price (₹ Lakhs)**

1 1000 2 50

2 1500 3 75

3 800 2 40

4 1200 3 60

5 2000 4 90

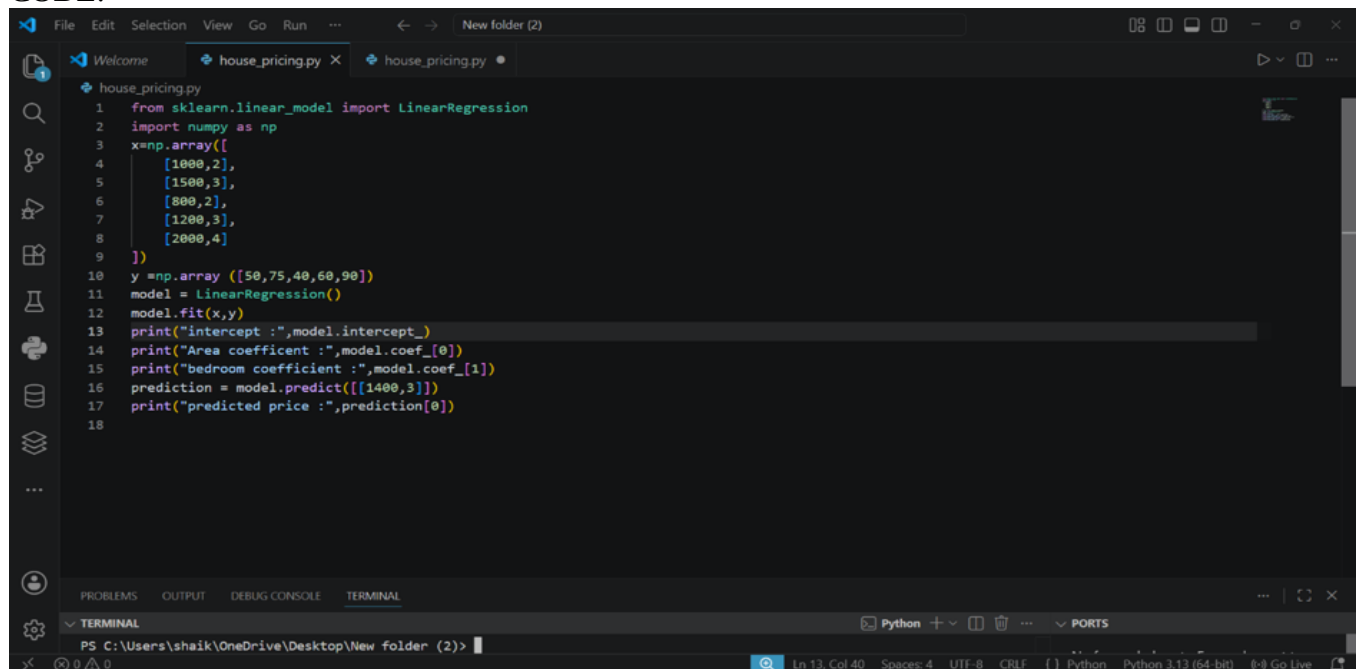
(i) Construct a linear regression model of the form

$$Price = \beta_0 + \beta_1(Area) + \beta_2(Bedrooms)$$

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

CODE:

A screenshot of a Python IDE (likely VS Code) with a dark theme. The editor window shows a file named 'house_pricing.py' with the following code:

```
1 from sklearn.linear_model import LinearRegression
2 import numpy as np
3 x=np.array([
4     [1000,2],
5     [1500,3],
6     [800,2],
7     [1200,3],
8     [2000,4]
9 ])
10 y=np.array ([50,75,40,60,90])
11 model = LinearRegression()
12 model.fit(x,y)
13 print("intercept :",model.intercept_)
14 print("Area coefficient :",model.coef_[0])
15 print("bedroom coefficient :",model.coef_[1])
16 prediction = model.predict([[1400,3]])
17 print("predicted price :",prediction[0])
18
```

The bottom of the IDE shows a terminal window with the command prompt 'PS C:\Users\shaik\OneDrive\Desktop\New folder (2)>' and the status bar at the bottom indicates 'Ln 13, Col 40', 'Spaces: 4', 'UTF-8', 'CRLF', and 'Python 3.13 (64-bit)'.

OUTPUT:

```
✓ TERMINAL

ers/shaik/OneDrive/Desktop/New folder (2)/house_pricing.py"
intercept : 8.50467289719628
Area coefficient : 0.0425233644859813
bedroom coefficient : -0.2803738317757003
predicted price : 67.196261682243
PS C:\Users\shaik\OneDrive\Desktop\New folder (2)> |
```

Q2. An email system uses the following feature values: (15 Marks - 20

Mins) **Email Word Frequency (Offer) Word Frequency (Win) Spam**

(1/0) 1 3 2 1

2 2 1 1

3 1 0 0

4 3 3 1

5 0 1 0

(i) Construct a linear classification model.

(ii) Determine the decision boundary.

(iii) Classify a new email with: Offer = 2, Win = 1.

CODE:

```
File Edit Selection View Go Run ... New folder (2)
email_spam.py house_pricing.py email_spam.py house_pricing.py
1 from sklearn.linear_model import LogisticRegression
2 import numpy as np
3 x = np.array([
4     [3,2],
5     [2,1],
6     [1,0],
7     [3,3],
8     [0,1]
9 ])
10
11 y = np.array([1,1,0,1,0])
12 model = LogisticRegression()
13 model.fit(x,y)
14 new_mail = [[2,1]]
15 prediction = model.predict(new_mail)
16 print("prediction: ",prediction[0])
17 if (prediction[0] == 1):
18     print("spam email")
19 }
20 else:
21     print("not a spam email")
22 }
23
24
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\shaik\OneDrive\Desktop\New folder (2)> |
```

OUTPUT:

TERMINAL

```
Users/shaik/OneDrive/Desktop/New folder (2)/email_spam.py"  
prediction: 1  
spam email  
PS C:\Users\shaik\OneDrive\Desktop\New folder (2)> 
```

Q3.A medical dataset is given:(15Marks -20 Mins)

Patient Fever (1/0) Headache (1/0) Disease (Yes/No)

1 1 1 Yes

2 1 0 Yes

3 0 1 No

4 1 1 Yes

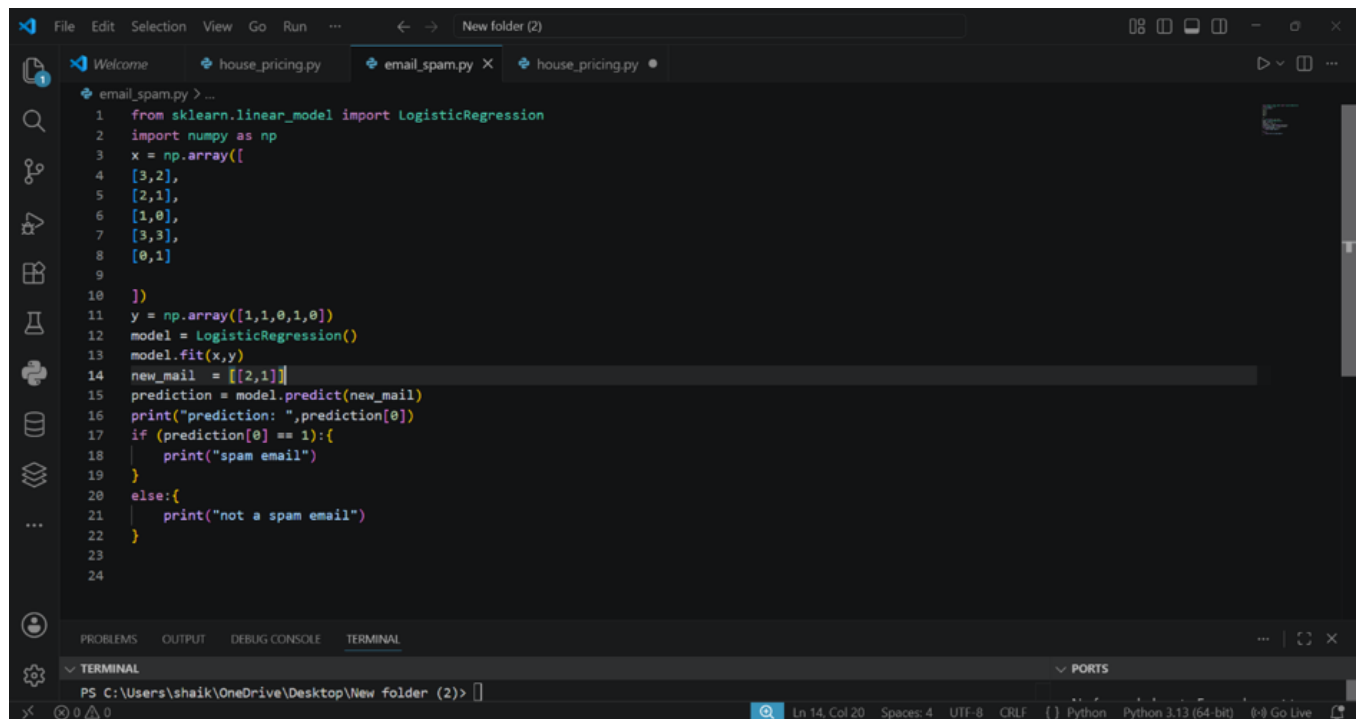
5 0 0 No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

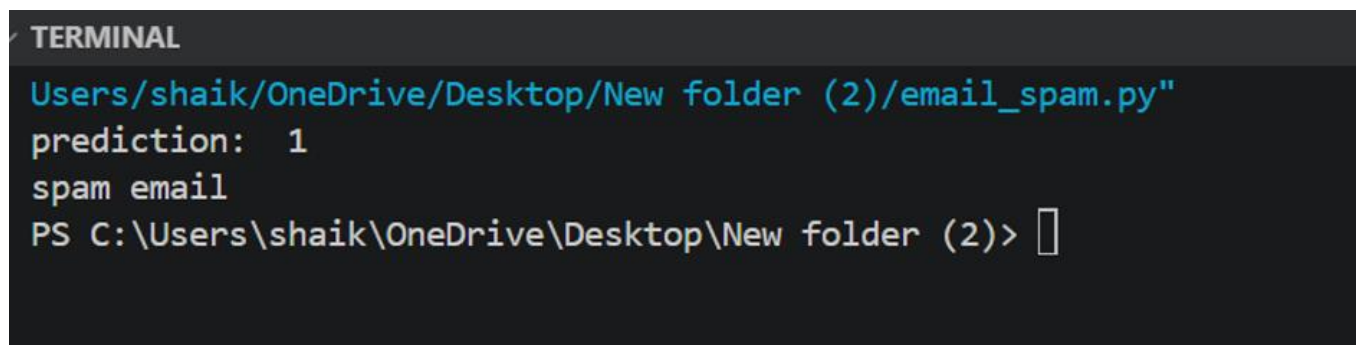
CODE:



The image shows a Visual Studio Code editor window with a file explorer on the left and a code editor in the center. The code editor displays a Python script named `email_spam.py`. The script imports `LogisticRegression` from `sklearn.linear_model` and `numpy` as `np`. It defines a feature matrix `x` and a target vector `y`, then fits a `LogisticRegression` model. A new email feature vector `new_mail` is defined, and the model's prediction is printed. An `if` statement checks the prediction; if it is 1, it prints "spam email", otherwise it prints "not a spam email". The bottom status bar shows the file is at line 14, column 20, with 4 spaces, UTF-8 encoding, and CRLF line endings. The Python interpreter is set to Python 3.13 (64-bit).

```
1 from sklearn.linear_model import LogisticRegression
2 import numpy as np
3 x = np.array([
4     [3,2],
5     [2,1],
6     [1,0],
7     [3,3],
8     [0,1]
9 ])
10
11 y = np.array([1,1,0,1,0])
12 model = LogisticRegression()
13 model.fit(x,y)
14 new_mail = [[2,1]]
15 prediction = model.predict(new_mail)
16 print("prediction: ",prediction[0])
17 if (prediction[0] == 1):{
18     print("spam email")
19 }
20 else:{
21     print("not a spam email")
22 }
23
24
```

OUTPUT:



The image shows a terminal window with a dark background. The title bar reads "TERMINAL". The prompt is `PS C:\Users\shaik\OneDrive\Desktop\New folder (2)>`. The output of the script is displayed in two lines: `prediction: 1` and `spam email`. The prompt is shown again at the end of the output.

```
Users/shaik/OneDrive/Desktop/New folder (2)/email_spam.py"
prediction: 1
spam email
PS C:\Users\shaik\OneDrive\Desktop\New folder (2)>
```

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
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Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
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